

Data logging. Data logging is very useful for a field job where you would want to log your measurements and recall them when you need. However, buy a multimeter with data logging facility only if you really need it because it will cause a big price difference.

Fast-response continuity latching. This is one of the features that you are going to use the most. Continuity testing is useful for tracing tracks, finding shorts and hundreds of other tasks. Fast-response continuity latching is a must for any multimeter. No datasheet mentions it but you need it as troubleshooting with a multimeter having slow-response continuity will be really annoying.

To assess a multimeter for fast response, set it in continuity mode and touch its probes to each other very quickly. Now do it faster and see the response. The tone should be solid and appear every time you touch the probes.

With fast-response continuity function you can find shorts on a chip just by wiping the probe across the pins, which is impossible with slow-response continuity multimeters.

Safety

This is the most important criterion as it involves your safety. There are several norms that a multimeter should comply with in order to assure safety. Otherwise, your multimeter might blow in your hand some day.

CAT rating. All handheld multimeter manufacturers are required to mention on their products the rated measurement category (CAT II, CAT III or CAT IV). This marking is a convenient way for users to identify the maximum transient voltage that a multimeter can safely withstand. Most



(L) Auto- and (R) manual-range multimeters

handheld digital multimeters have this rating marked near the voltage/current input terminals. Most multimeters are CAT III rated but beware of some cheap Chinese meters that have CAT III written on them just because users are looking for it.

CAT III. This rating applies to building circuit installations that are completely within the building, including parts of the service panel and branch circuits. It also applies to many of the building's fixed equipment, which are connected directly to the building's mains supply instead of being connected through cords and plugs.

Fuse rating. The datasheet of a cheap multimeter may mention that the multimeter complies with all the norms but it really may not. Check the fuse ratings for both current ranges. To be sure, open the cabinet and check the type and rating of the fuse yourself before directly measuring mains through it. It is highly dangerous. Never buy a cheap multimeter for mains measurements.

Shrouded input plug. Choose a multimeter with shrouded input plug so that you do not get a shock while doing mains measurement.

Good-quality probes.

Probes should be of very good quality for high-voltage measurements. Some manufacturers do not properly insulate the probes. Measuring mains supply with such probes is very dangerous. Look for high-voltage silicon insulation.

Construction and build

This is the last evaluation criterion. It mainly depends on your work environment requirements. Though exact requirements will depend on your specific work environment, here are some basic things to look for:

Separate fuse and battery compartments. Not all multimeters have a separate fuse and battery compartment, but it will be really helpful if your multimeter features one. A separate compartment will allow easy removal for replacement of batteries and fuse. It will also mean no major damage in case the fuse blows.

Rugged input jacks. Input jacks should be rugged from inside because you will be taking out the probes and putting them back quite frequently. There is a high chance of the jack becoming loose, resulting in improper measurements.

Rugged build. For field use, it is good to have a multimeter with rugged build. It will also help the multimeter withstand any accidental fall. Normally, multimeters are provided with rubber covers that work as shock absorbers to help protect the multimeter from any damage. ●

The author is technical editor at EYF



Subscribe to **Electronics For You**
For 3 Years And GET an
EFY-KnS PIC Development
Board worth Rs 1225/-, FREE



An 8-Bit Microcontroller PIC Development Board (16F8771) Which can act as a PIC Programmer too. A Perfect tool to develop applications around PIC Microcontrollers. Easy to test and debug since LCD module & LEDs are already integrated on the Board. High Flexibility provided in terms in Using Input/Output Ports. The board can be powered through USB port as well as DC adapter.

To BUY Call: 08800100761 or
Visit pay.efyindia.com/freekits to BUY ONLINE!

EFY Enterprises Pvt Ltd, D-87/1, Okhla Industrial Area, Phase 1, New Delhi 110020 • Phone: 011-26810601/02/03